

PhD student in Cybersecurity at RIT working on LLM security (inference-cost / resource-exhaustion stressors) and software security analysis, including side-channel and non-constant-time (non-CT) verification. Previously fullstack and SIEM engineering at Logpoint.

## Work Experience

### Rochester Institute of Technology

Ph.D. Research (Advisor: Yinxi Liu)

Aug 2025 – Present

Rochester, NY

- LLM security: inference-cost / resource-exhaustion stressors and related measurement studies (RASE'26); metamorphic stress testing of vision-language components for time-critical systems (TRUST'26).
- Software security analysis focused on side-channel and non-constant-time (non-CT) verification for cryptographic and systems code (ongoing research).

### Logpoint Inc.

Software Engineer

April 2024 – July 2025

Lalitpur, Nepal

- Built and maintained Linux application-server monitoring tools (Linux, Go, Python, TypeScript, GraphQL).
- Owned the License Portal lifecycle for packaging and distributing customer licenses (Rust, TypeScript, Node.js, gRPC, Ansible, GitLab CI, Docker).
- Contributed to AgentX EDR; fixed bugs and performance issues.

### Logpoint Inc. / Tribhuvan University

Lab Tutor (Part time)

Jan 2024 – March 2024

Lalitpur, Nepal

- Assisted lecturer in “Security Operations Fundamentals” (Logpoint + Pulchowk Campus collaboration).
- Helped create course materials, evaluate assignments, and run labs on phishing awareness, threat modeling, and SIEM.
- Guided labs on network intrusion, web and Linux security, and application security using [pwn.college](https://pwn.college).

### Logpoint Inc.

Solutions Engineer

May 2023 – March 2024

Lalitpur, Nepal

- Handled L3 support for operating systems, application, disk, and network issues.
- Resolved 150+ customer cases; conducted technical training for new hires.
- Worked with SSL/TLS, PKI, and AWS.

### Logpoint Inc.

Associate Solutions Engineer

April 2022 – April 2023

Lalitpur, Nepal

- Debugged, tested, and reviewed SIEM application components for stability and performance.
- Resolved Linux and Windows issues related to CPU, memory, and network.
- Worked with cryptography, authentication/authorization, REST APIs, cloud, Nxlog, and Wazuh.
- Troubleshoot issues across Python, Java, SQL, MongoDB, Docker, and Kafka; 450+ cases with RCA.

### Techniti Nepal

Software Developer and Cofounder

Feb 2021 – March 2022

Lalitpur, Nepal

- Built web applications for healthcare, management portals, and a cricket fantasy platform.
- Worked with Python, JavaScript, MySQL, Linux servers, Docker, and GitHub Actions.
- Participated in the full software development lifecycle and helped keep applications available under load.

### Sireto Technology

Software Engineering Intern (part-time)

Sept 2021 – Dec 2021

Lalitpur, Nepal

- Worked on Haskell smart contracts on the Cardano platform.
- Completed the Plutus Pioneer Program.

## Education

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### Ph.D. in Cybersecurity

Aug 2025 – Present

Rochester Institute of Technology (RIT), Golisano College of Computing and Information Sciences Rochester, NY

- Advisor: Yinxi Liu ([yxlics@rit.edu](mailto:yxlics@rit.edu)).
- Focus: LLM security (inference-cost stressors) and software security analysis (side-channel / non-CT verification).

### Bachelor in Computer Engineering

November 2017 – April 2022

Pulchowk Campus, Institute of Engineering, Tribhuvan University

Lalitpur, Nepal

- Capstone Project: Speech Recognition In Nepali Language.
- Academic Supervisor: Subarna Shakya, Ph.D.
- Entrance exam rank 29 / ~18,000 (2017); First Division, 78.21% aggregate.

### High School

June 2015 – June 2017

Liverpool International College

Kathmandu, Nepal

- Graduated in the top 5 of the class; 85.8% aggregate.

## Publications

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### Conference Paper

Arman Chhetri, Yinxi Liu. “Metamorphic Stress Testing of Vision-Language Components for Time-Critical Autonomous Systems.” *TRUST’26*, co-located with ASE 2026, Munich, Germany, Oct 12–16, 2026. Accepted 2026-08-23. DOI: [10.1145/3843782.3844660](https://doi.org/10.1145/3843782.3844660).

### Conference Paper

Arman Chhetri, Yinxi Liu. “When Do Inference-Cost Stress Tests Generalize? A Cross-Model and Cross-Configuration Measurement Study.” *RASE’26*, co-located with ASE 2026, Munich, Germany, Oct 12–16, 2026. Accepted 2026-08-25. DOI: [10.1145/3820756.3844910](https://doi.org/10.1145/3820756.3844910).

### Conference Paper

M. Dhakal, A. Chhetri, A. K. Gupta, P. Lamichhane, S. Pandey and S. Shakya, “Automatic speech recognition for the Nepali language using CNN, bidirectional LSTM and ResNet,” 2022 International Conference on Inventive Computation Technologies (ICICT), Nepal, 2022, pp. 515–521, doi: [10.1109/ICICT54344.2022.9850832](https://doi.org/10.1109/ICICT54344.2022.9850832).

## Select Projects

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### ContamBench

Dataset and CLIs for contamination stress testing of open-weight LLM serving

- Builds contamination-oriented stress cases and evaluation CLIs for open-weight LLM serving stacks.
- Supports measurement of inference-cost / resource-pressure behavior under contaminated prompts.

### Build Git from scratch

Small version of GIT implemented from scratch (Go)

- Implements core Git-style version-control operations from first principles.
- Written in Go as a learning-focused recreation of essential Git internals.

### Distributed Filestore

Flat file store distributed across multiple nodes (Go)

- Stores and retrieves files across multiple nodes in a flat distributed layout.
- Written in Go for simple multi-node file distribution.

## Interpreter from scratch

*Lox interpreter implemented from scratch (Rust)*

- Implements a Lox interpreter from scratch in Rust.
- Covers core language pipeline pieces (lexing, parsing, and evaluation).

## Technical Skills

- Research: LLM inference tooling (vLLM), experimental evaluation / stress testing, software security analysis (side-channel / non-CT verification)
- Languages: Python, Go, Rust, TypeScript/JavaScript, Bash, C; exposure to Haskell
- Frameworks & APIs: FastAPI, NestJS, GraphQL, gRPC, REST; prior Vue.js / Flask
- Systems: Linux administration, Docker, networking, system troubleshooting
- DevOps / industry: Ansible, GitLab CI / GitHub Actions, SIEM operations (prior Logpoint)
- Troubleshooting tools: netstat, Wireshark, strace, perf

## Soft Skills

- Clear written and spoken English communication
- Problem solving and debugging under time pressure
- Fast learner with strong research habits
- Collaborative work across research and engineering teams

## License and Certifications

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|---------------------------------|----------------------------------|
| <b>ISC2</b>                     | Certified in Cybersecurity       |
| <b>LinkedIn</b>                 | Advanced Linux: The Linux Kernel |
| <b>Cisco Networking Academy</b> | Networking Essentials            |
| <b>AWS</b>                      | AWS Academy Graduate             |
| <b>Coursera</b>                 | Algorithmic Toolbox              |
| <b>Coursera</b>                 | Deep Learning Specialization     |

## Affiliations

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|------------------|-------------------------------------------------------------|-----------|
| <b>LOCUS</b>     | Datarush Contest Organizer                                  | 2021      |
| <b>LOCUS</b>     | Children In Tech Organizer                                  | 2021      |
| <b>Microsoft</b> | Microsoft Learn Student Ambassadors                         | 2019      |
| <b>ANSWER</b>    | American Nepal Student and Women Education Relief's Scholar | 2012–2022 |

## References

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|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Prof. Dr. Subarna Shakya</b> | Professor of Computer Engineering, Department of Electronics and Computer Engineering, Pulchowk Campus, Institute of Engineering, Tribhuvan University, Email: <b>drss@ioe.edu.np</b> |
| <b>Santosh Baral</b>            | Director of Support and Operations, Logpoint Inc. Lalitpur, Email: <b>sab@logpoint.com</b>                                                                                            |